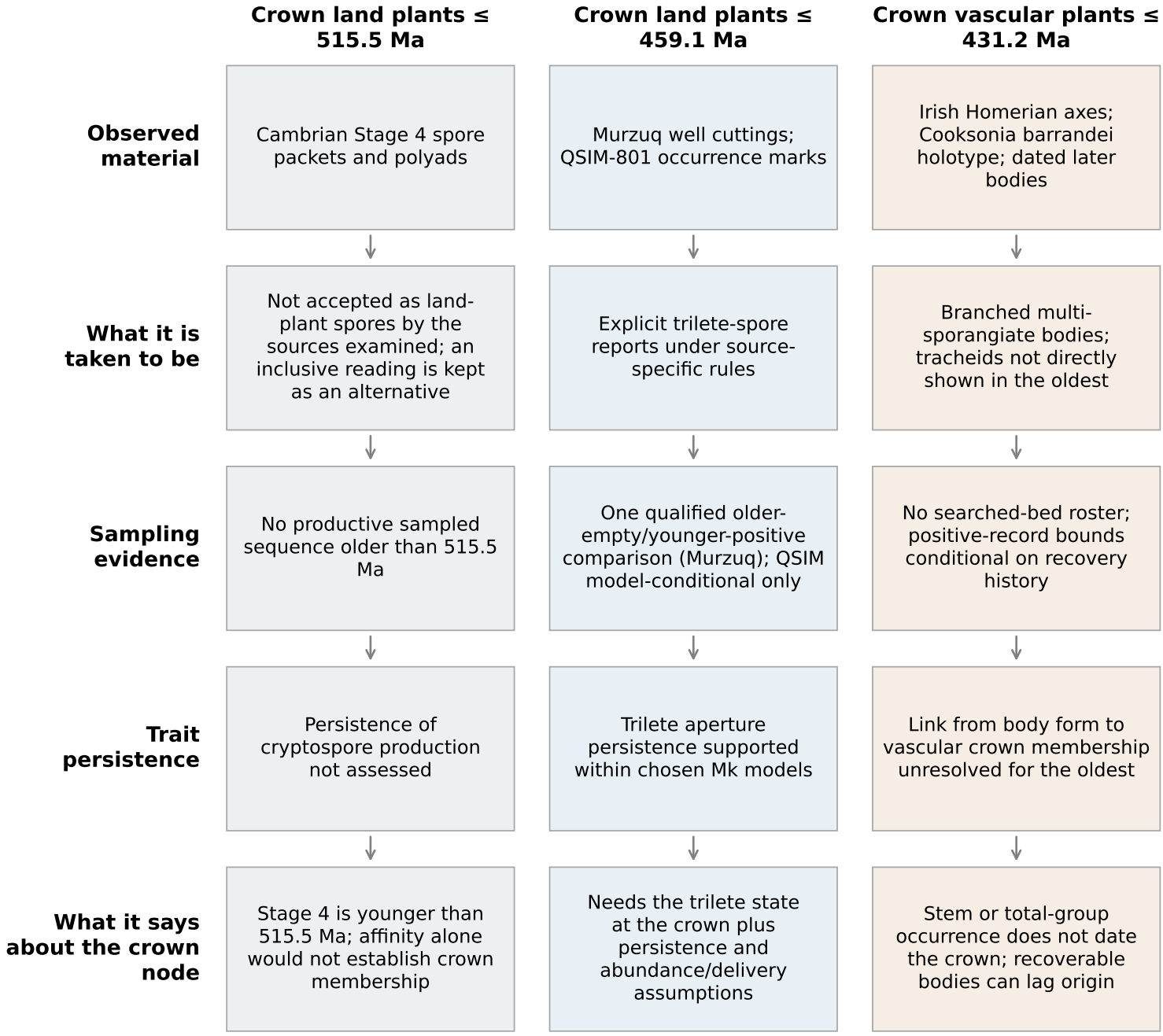
Crown land plants ≤ 515.5 Ma
Crown land plants ≤ 459.1 Ma
Crown vascular plants ≤ 431.2 Ma
Observed material
Cambrian Stage 4 spore packets and polyads
Murzuq well cuttings; QSIM-801 occurrence marks
Irish Homerian axes; Cooksonia barrandei holotype; dated later bodies
What it is taken to be
Not accepted as land-plant spores by the sources examined; an inclusive reading is kept as an alternative
Explicit trilete-spore reports under source-specific rules
Branched multi-sporangiate bodies; tracheids not directly shown in the oldest
Sampling evidence
No productive sampled sequence older than 515.5 Ma
One qualified older-empty/younger-positive comparison (Murzuq); QSIM model-conditional only
No searched-bed roster; positive-record bounds conditional on recovery history
Trait persistence
Persistence of cryptospore production not assessed
Trilete aperture persistence supported within chosen Mk models
Link from body form to vascular crown membership unresolved for the oldest
What it says about the crown node
Stage 4 is younger than 515.5 Ma; affinity alone would not establish crown membership
Needs the trilete state at the crown plus persistence and abundance/delivery assumptions
Stem or total-group occurrence does not date the crown; recoverable bodies can lag origin